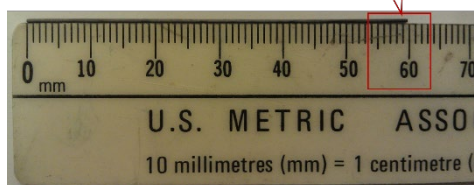


## Significant Figures in Calculations, Unit Conversion, and Unit Tracking

**Synopsis:** Remember, the number of significant figures indicates the measurement precision. If you use a ruler with the smallest markings at 1 mm intervals, your ability to measure a distance is limited to  $\pm 0.5$  mm precision. For example, a ruler is measuring the length of the straight black line. The end of the line falls somewhere between 59 and 60 mm. With this ruler, the acceptable values are 59.5 mm, 59.8 mm, and 60.0 mm, and so on. That is somewhere between 59.5 and 60.0 mm is acceptable.



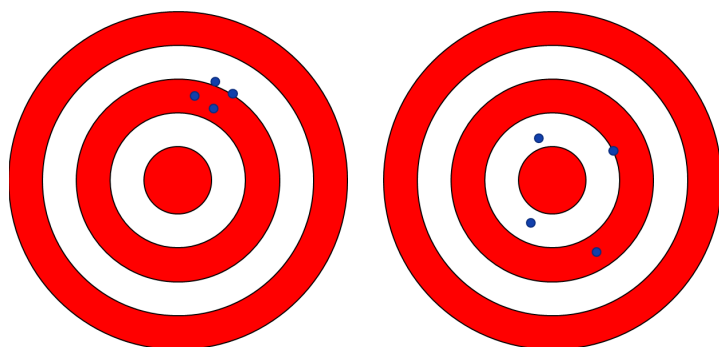
The takeaway message is this: When a quantity falls between the smallest interval, you must estimate the digit beyond the smallest scale. Here, you "guess" that to the first decimal place ( $10^{\text{th}}$  of an mm), the length can be 59.5 mm to 60.0 mm, where the first decimal is your estimation.



Finally, you must indicate the range of uncertainty, which is half of the smallest interval:  $59.8 \pm 0.5$  mm.

This uncertainty indicates the degree of error in your measurement, which is generally given as the % error:

$$\%error_{\text{Precision}} = \frac{\text{uncertainty in measurement}}{\text{measurement}} \times 100 = \frac{\delta X}{X} \times 100$$
, where the uncertainty in measurement is  $\delta X$  and the measurement is  $X$ . In the above example,  $X$  is the length of the straight black line  $X = 59.8$  mm,  $\delta X = 0.5$  mm, so the % error is  $(0.5/59.8) \times 100 = 0.8\%$ . The % error is given as an absolute value (always positive). You may find this type of error denoted as "precision error."

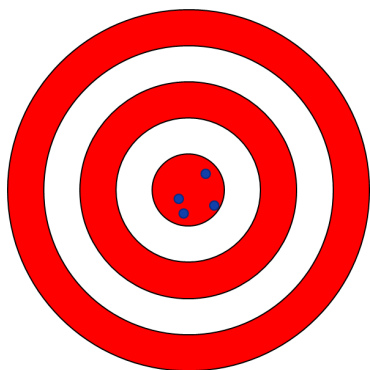


Precise

Accurate

The other type of error we will discuss is the accuracy error. But first, what is the difference between "precision" and "accuracy"? See left and note what you think is the difference.

See below for both a precise and accurate set of data points. What do you notice?



Precise  
Accurate

To denote accuracy error in percentage terms, we do the following:

$$\%error_{accuracy} = \frac{|\text{Expected} - \text{Measured}|}{\text{Expected}} \times 100$$

. We typically mean "Expected" as in theoretical prediction. "Measured" is what you actually observed. Going forward, in future labs, you will most likely just have to deal with accuracy error in percent. %error can simply be a comparison between two sets of measurements where the difference is not expected.

For example, a theory predicts that the acceleration due to gravity on the surface of the earth is  $9.8 \text{ m/s}^2$ . You do a set of measurements and determine

that the observed value is  $10.2 \text{ m/s}^2$ . The %error is  $\frac{|9.8 - 10.2|}{9.8} \times 100 = 4.1\%$ .  
Note that % error is given as an absolute value (always positive).

Next, we must discuss what happens when you add, subtract, multiply, and divide numbers with a fixed number of significant figures.

Imagine you have two length measurements: a field's width (W) and length (L).  $W = 59.893 \text{ m}$  and  $L = 34.32 \text{ m}$ . I want to know the area (A) and perimeter (D).

$A = W \times L$  and  $D = 2 \times (W + L)$ . How should we handle the math operations? Notice that the width is somehow measured with a higher precision than the length (which can happen if you use a different device/method to do the measurement, for whatever reason).

Here is the rule of thumb: Your estimations of area and perimeter will be as precise as the least precise number. Here, because your length is less precise, its precision will determine the precision of the area and perimeter.

Here are the simple rules. 1) Multiplication and division: The estimated value will have a number of significant digits equal to the smallest number of significant figures. In the above example, the area will have 4 sig. fig. since length only has 4 sig. fig. 2) Addition and subtraction: The estimated value will have the decimal place (or digit) the same as the least precise number. In our example, the perimeter will only go to the second decimal place.

$A = 59.893 \text{ m} \times 34.32 \text{ m} = 2055.52267 \text{ m}^2 = 2056 \text{ m}^2$  to 4 s. f. Note that the last sig. fig., 6, is due to rounding up of the next digit (2055.5 rounded up to 2056 to 4 s. f.). This is the standard practice.

$D = 2(59.893 \text{ m} + 34.32 \text{ m}) = 2(94.214) = 188.426 \text{ m} = 188.43 \text{ m}$  to 2 decimal places. Note, here also, that the last digit requires looking at one digit beyond and rounding up.

### Unit Conversion:

$$\# \text{ Desired unit} = \# \text{ Starting unit} \times \left( \text{Conversion factor} \times \frac{\text{Desired unit}}{\text{Starting unit}} \right), \text{ where } \# \text{ is the numerical value.}$$

**e.g., convert 1.76 miles to m. (# starting unit) = 1.76 miles. Desired unit = m. The conversion factor must be carefully chosen using 1 mile = 1609.34 m (Google it).**

$$\# \text{ Desired unit} = 1.76 \text{ miles} \times \left( \frac{1609.34 \text{ m}}{1 \text{ miles}} \right)$$

$$1.76 \text{ miles} \times \left( \frac{1609.34 \text{ m}}{1 \text{ miles}} \right) = 2830 \text{ m} \quad \text{to 3 s.f.}$$

### **Unit Tracking:**

**F = ma, where F is the force in newton (N), m is mass in kg, and a is acceleration in m/s<sup>2</sup>. What is the unit of force, N, in terms of kg, m, and s (i.e., standard units of mass, length, and time)?**

$$F = ma$$

$$[N] = [kg] \times [m / s^2]$$

**Distance is given by velocity × time. Confirm this formula is correct by tracking the unit.**

**Time is given by distance/velocity. Confirm that units track correctly.**

**Work is measured in joules (J) and is given by force × distance. If force is measured in N, distance in m, what is the unit of work (J) in terms of standard unit?**

1. Write each number in scientific notation with the correct number of significant figures.

a. 9624 \_\_\_\_\_

b. 0.82 \_\_\_\_\_

c. 500.06 \_\_\_\_\_

d. 620.0 \_\_\_\_\_

2. Calculate the following to the appropriate number of significant figures. All numbers are found from measurement. Show all work.

a.  $18.998403 + 83.80 =$

b.  $\frac{34.60}{0.00715} =$

c.  $(1.36 \times 10^{-4}) + (3.11 \times 10^{-5})$

d.  $41,600 - 213.1$

e.  $\frac{6.028 + 94.17}{6.13 - 2.068} =$

f.  $(427.932 / 396.5) - (0.07461 / 0.082967) =$

3. Convert 2.75 m to each of the following:
- a. mm

b. nm

c. km

d.  $\mu\text{m}$

4. Convert 3.4 mg to each of the following:

a. ng

b. kg

c. cg

d. Mg

5. Convert 5.7 hours to each of the following:

a. s

b. ms

c. Ms

d. ps